Final Project Instructions

MAE 341: Spaceflight
Part 1 Due Wednesday, November 26, 2025
Part 2 Due Friday, December 12, 2025 (Dean's Date)

Final Project: Model an Existing Mission Concept in STK & Build On It

In this project, for **Part 1**, you will model an existing space mission concept in **STK** and write a formal report explaining the mission design, trajectory, and purpose. Your chosen mission may be real or fictional, but it must have enough publicly available information to be modeled in STK at a reasonable level of detail. In **Part 2**, you will build on your Part 1 mission with your own mission design. Propose additional phases, alternate objectives, or new operations; then implement and analyze your design in STK. Submit a single PDF report (Overleaf/LaTeX strongly encouraged) and a link to a video of your STK simulation as your deliverable.

Example Missions to Model

You may choose from the following ideas or propose your own mission concept. If you are unsure whether your proposed concept is appropriate, ask the course staff.

- The first Moon landing (**Apollo 11**)
- The **Artemis I** mission (November 2022) — uncrewed flight test of Orion around the Moon, returned to Earth
- The **Artemis II** mission (planned 2026) — first crewed lunar flyby of the Artemis program
- Assembly steps of the **International Space Station (ISS)**
- The planned deorbiting of the ISS
- A **LEO mission** with specific tracking or observation goals
- The **Rich Purnell Maneuver** from *The Martian*
- **NASA / Princeton's IBEX** spacecraft — features a complex lunar resonance orbit
- **Mars rover landings** (select one in detail, or compare multiple)

Report Structure

Your report should follow a professional format, similar to Projects 1 and 2, but with additional emphasis on mission design and clarity. Think of this as a mini **NASA Preliminary Design Review (PDR)** summary — concise but complete.

Part A: Replicate an Existing Mission

Purpose: Demonstrate understanding of an existing mission and reproduce its essential behavior in STK.

Mission Goals and Objectives State the mission goal (high-level purpose) and list specific, measurable objectives (e.g., *achieve lunar flyby at* $r_p = \cdots$, *demonstrate reentry corridor*, *obtain imaging coverage over region* X). Include mission type (scientific, exploration, communications, technology demo, etc.).

Mission Concept Overview Brief background (who, when, why), major phases (launch, cruise, insertion, operations, return), and any key milestones.

STK Model Include the link to a video of your STK simulation. A screen-recording of the standard STK simulation view is sufficient. In the PDF, include enough images and relevant graphs for your mission from STK.

STK Modeling Setup Briefly explain what you did in STK to create your project. Include things like scenario timeframe, objects, propagators, force models (e.g., HPOP), frames, and assumptions. Explain any STK tools used (access, coverage, sensors, comms). Integrate labeled screenshots/plots throughout to aid in your explanation.

Results and Analysis Show ground tracks, altitude/time plots, access windows, maneuver timelines, etc. Explain how these support that your STK replication captures the mission's overall behavior. Note limitations and what finer-fidelity items you would add with more time.

Discussion and Interpretation What did you learn about the mission's dynamics and trade-offs? How do your results relate to the real mission?

Part B: Extend with Your Own Design

Purpose: Build on Part A with an original, technically justified mission extension.

Mission Goals and Objectives (Extended) Define an expanded goal and specific, measurable objectives for your extension (e.g., add lunar orbit insertion and 6.5-day surface sortie; introduce a cislunar relay; perform autonomous stationkeeping; add abort/contingency options). Provide a clear statement of mission type.

Mission Concept and Design Rationale Explain what your spacecraft/system is doing in the extension, the problem it solves or capability it demonstrates, and why your chosen orbits/configuration/sequence are appropriate. Outline phases: launch (if changed), transfers, operations, return/deorbit.

Updated STK Model Include the link to a video of your updated STK simulation. A screen-recording of the standard STK simulation view is sufficient. In the PDF, include enough images and relevant graphs for your mission from STK. Be clear about what part is your original contribution to the mission concept.

Updated STK Modeling Setup Explain what you did in STK to create your modified mission. Describe new objects, updated propagators/force models, time frames, and assumptions. Identify the STK tools used for the extension (coverage, access, comms, sensors, targeting). Include labeled visuals.

Results and Analysis (Extended) Present the key figures that demonstrate feasibility and performance (*e.g.*, LOI timing and altitude, access to ground stations, coverage over a target region, delta v budgets as estimated via STK or supporting calculations). Discuss limitations (idealized orbits, simplified dynamics, notional subsystems).

Discussion and Interpretation (Extended) Reflect on what changed relative to Part 1, what worked, and what required design trade-offs. Comment on realism: what would be required to make it operational?

Conclusions

Summarize the principal findings from Parts A and B, and the key takeaways regarding feasibility, risks, and recommended next steps.

Formatting and Submission Expectations

- Write in full paragraphs.
- Use **labeled figures** and reference them in your text (e.g., "As shown in Figure 3, the lunar flyby occurs at t = 72 hours").
- Integrate visuals throughout — do not append unlabeled screenshots.
- Keep your analysis concise: about 1–2 paragraphs per major phase or section.
- The main emphasis of this project is on your **STK modeling**. Include enough screenshots and explanations of your modeling methods to demonstrate your work clearly.
- Properly cite all external resources or data you use.

Rubric (100 points total)

Grading emphasizes both accurate replication and thoughtful extension.

- **Part 1: Understanding of chosen mission** — 25 pts
- **Part 1: STK modeling quality/correctness** — 25 pts
- **Part 2: Mission design quality/originality** — 25 pts
- **Part 2: STK modeling accuracy/completeness** — 15 pts
- **Clarity, organization, and use of visuals** — 10 pts

Research and References

It is your responsibility to find the information necessary to inform your STK modeling. The **NASA Technical Reports Server (NTRS)** is a great resource for this. See: https://ntrs.nasa.gov/. For example, here is an example of a helpful paper on the *Rich Purnell Maneuver* from *The Martian*: https://ntrs.nasa.gov/api/citations/20150019662/downloads/20150019662.pdf

Online STK tutorials found on the Ansys website (which you likely used for Project 2) will also be a helpful resource for this project.

Existing STK projects by other people are available online, such as at https://astrogatorsguild.com/the-astrogators-guild/. You are welcome to look at such projects for inspiration, but your submitted project must be your own STK scenario started from a blank scenario file.

Generative AI should not be used to write the report for you. Acceptable uses of AI include finding technical papers, formatting into LaTeX, and troubleshooting STK. Everything submitted in Project 3 must be your own work. Please disclose your use of Generative AI in the references section of your report.

General Advice

Each of these examples can become quite detailed, so you will need to balance modeling complexity and analysis depth:

- If you select a **complex mission** (e.g., IBEX or a multi-phase Artemis trajectory), it's fine to keep the modeling higher level and focus on key phases and results.
- If you select a **simpler mission** (e.g., a single-satellite LEO mission), include additional STK detail such as communications, sensor modeling, or access analysis.

Use your judgment to maintain a reasonable scope while still demonstrating that you understand the mission's goals and how it operates.

You need to use your judgment on what information is appropriate to include for your project in Results and Analysis. Use your Mission Goals and Objectives section to identify a metric with which to assess your model. For example, if you have a LEO sensing satellite, use STK to get a quantitative metric for your sensing capabilities. If you are modeling the Rich Purnell maneuver, you could investigate propellant/ delta V requirements and to what extent the maneuver is feasible. TL;DR: Pick a measurable parameter (ground coverage, delta V, time), assess it, and explain its impact on your mission.

Please post your questions about the project on Ed Discussions or attend office hours.